

INTRODUCTION

XLSTAT makes accessible to anyone a powerful, complete and user-friendly data analysis and statistical solution. Accessibility to the XLSTAT interface totally relies on Microsoft Excel, whether for inputting the data or for displaying the results. The power of XLSTAT is a result of all computations having been developed with the C++ programming language. Users' suggestions have led to software improvements that offer a complete solution to a well-adapted variety of requirements. The user-friendly interface facilitates the use of some statistical methods that might require hours of training with other software.

More detail about the statistical processes described in this card can be found in the Help option available on the XLSTAT ribbon.

COMMON DIALOG BOX TOOLS



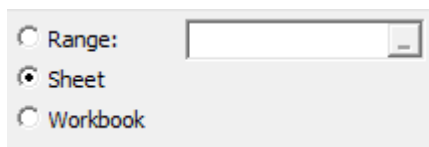
Click to reset the dialog box to default options.



Click to delete existing data selections of the dialog box.



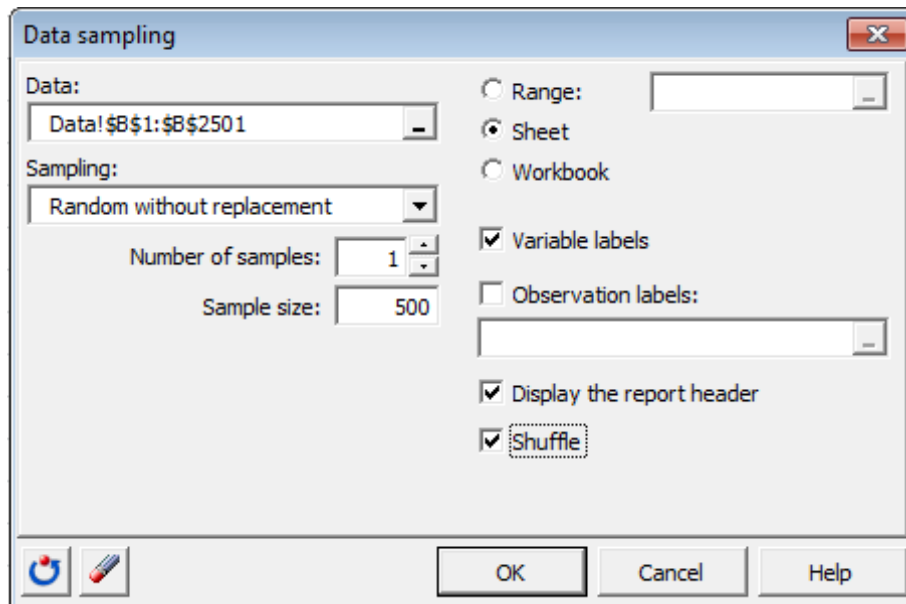
The blue arrow indicates that data are displayed in columns; change the arrow to red to indicate that data are displayed by rows.



You may choose to locate the generated subsample beginning in a cell in the existing worksheet (Range), in a new worksheet within the current workbook (Sheet), or in a new workbook (Workbook).

DATA SAMPLING

Use the Data Sampling tool found in the Preparing data dropdown object to generate a subsample of observations from a dataset.



Select the data in the Excel worksheet, choose the sampling method, and enter the sample size to be generated. You may choose to locate the generated subsample beginning in a cell in the existing worksheet, in a new worksheet within the current workbook, or in a new workbook. The variable labels checkbox is selected if the first row of the data contains a data label (header). Default settings indicate that a report header will be displayed and that the sample output data will automatically be permuted. Uncheck these options if no report header is required and if you want to display the data in the order that it was generated.

Sample report header associated with using the Random without replacement sampling method to generate a sample of 500 from

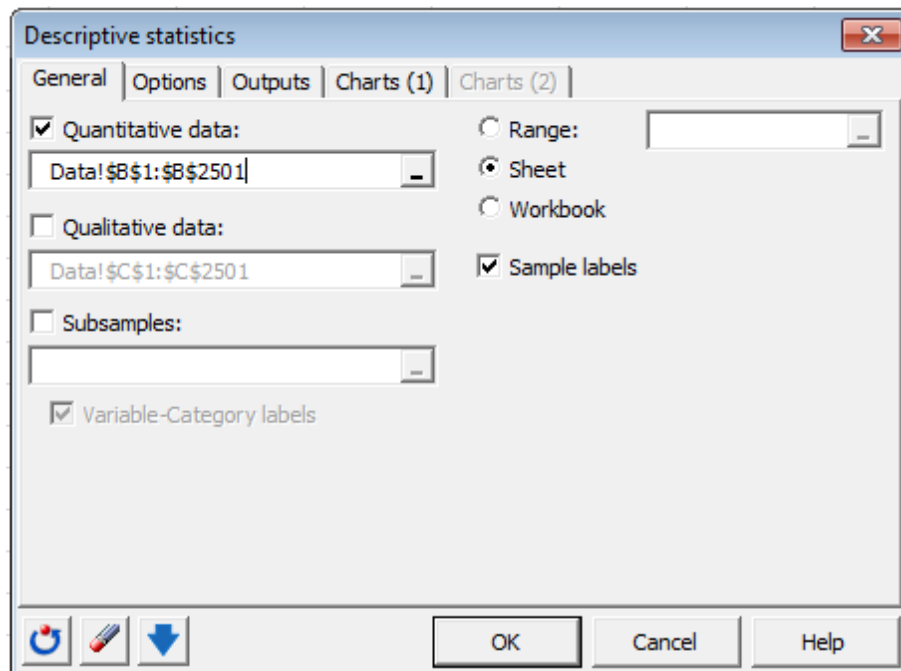
XLSTAT									
2015.2.0									
Data: Workbook = EAI.xlsx / Sheet = Data / Range = Data!\$B\$1:\$B\$2501 / 2500 rows and 1 column									
Sampling method: Random without replacement									
Number of samples: 1									
Sample size: 500									
Shuffle: Yes									
Seed (random numbers): 4331250									
the given dataset.									

DESCRIPTIVE STATISTICS

The Descriptive Statistics tool (found in the Describing data tab) is used to calculate descriptive statistics and display univariate plots for quantitative and/or qualitative variables.

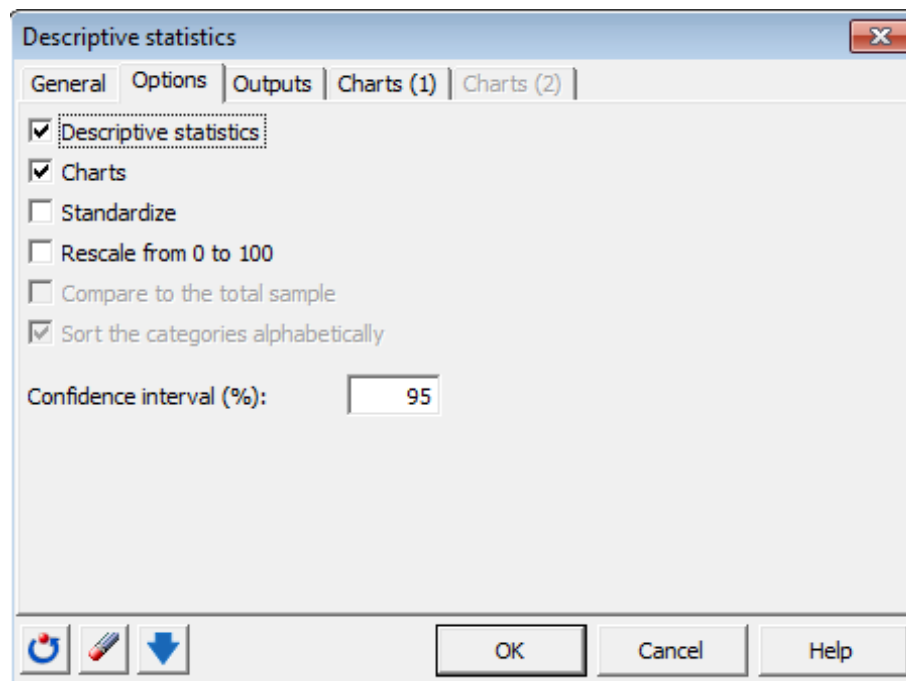
GENERAL TAB

In the General tab, sample data is selected and the report output location is specified.



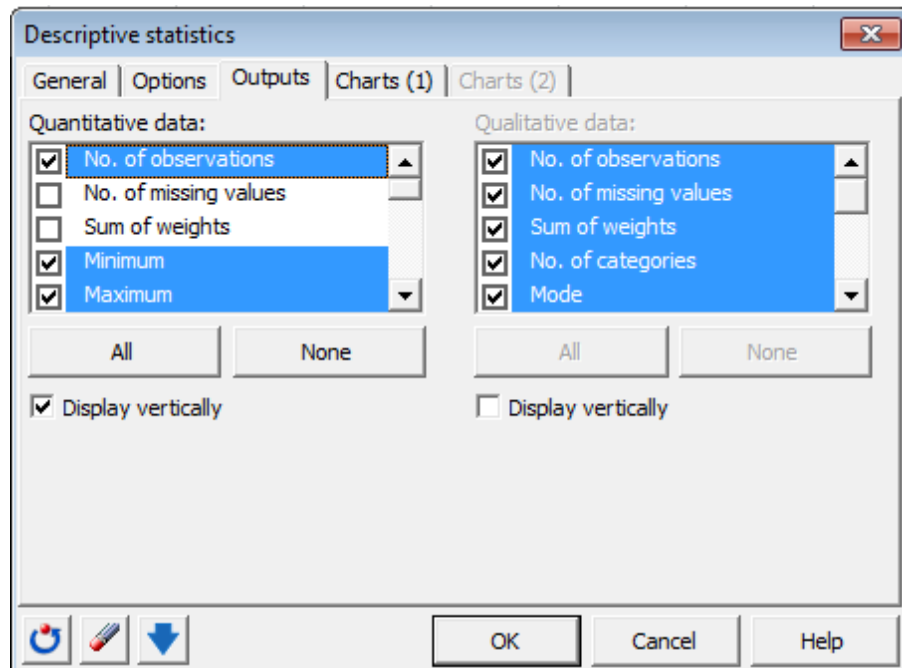
OPTIONS TAB

The Options tab provides options for descriptive statistics, charts, normalization and scaling of data, and specification of a confidence interval.



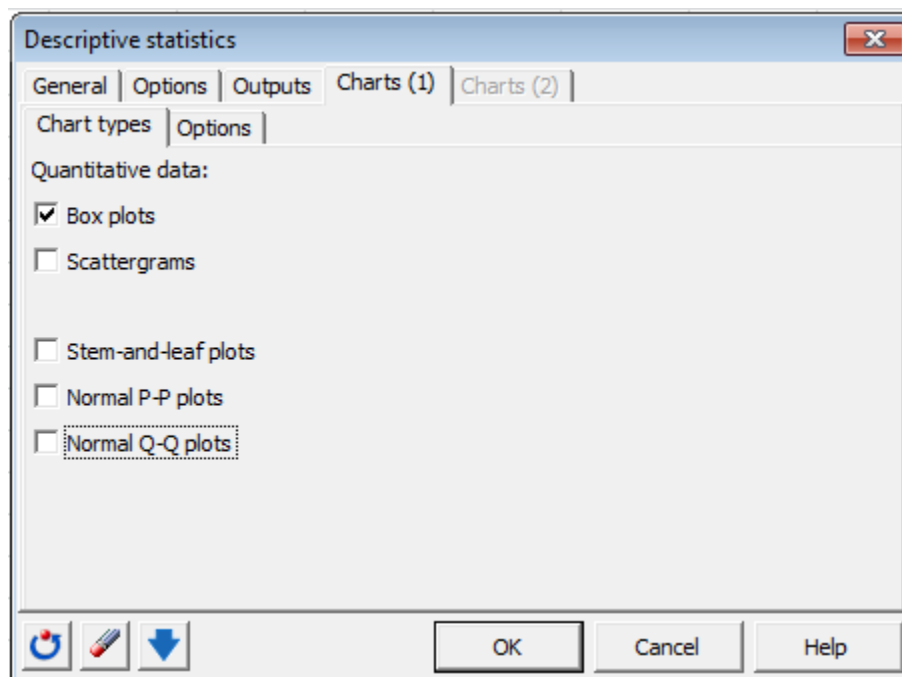
OUTPUTS TAB

The Outputs tab allows you to choose the specific options you want to calculate for quantitative and/or qualitative data.



CHARTS(1) TAB

The Charts(1) tab specifies chart types for quantitative data including box plots, scatter grams, stem-and-leaf plots, P-P plots, and Q-Q charts.



CHARTS(2) TAB

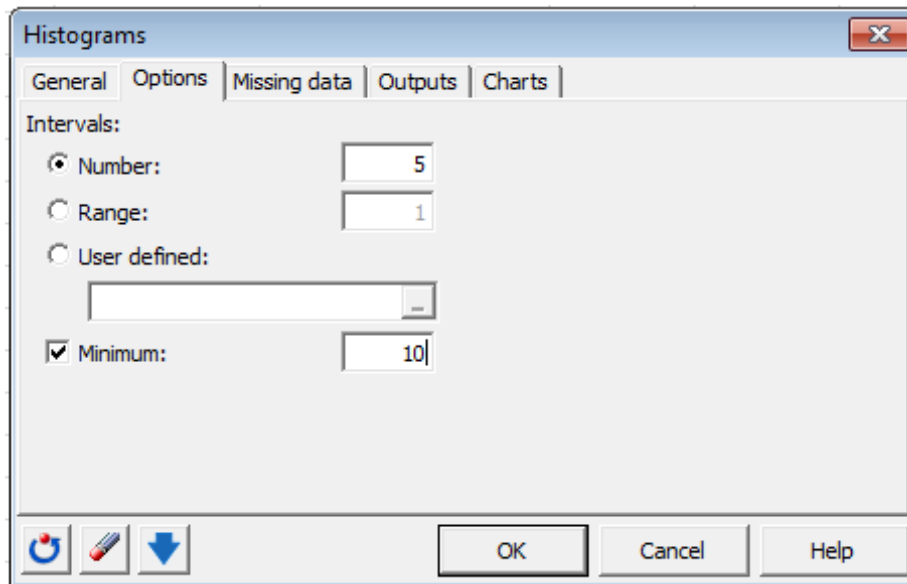
The Charts(2) tab provides charting options for qualitative data including bar charts, pie charts, doughnuts, stacked bars, multiple bars. In this tab you may choose to plot frequencies or relative frequencies of the categories. This tab becomes available when the Qualitative data option is selected in the General tab.

HISTOGRAMS

Histograms – Use this tool (found in the Visualizing and Describing data tabs) to create a histogram from a sample of continuous or discrete quantitative data. Data and data type are selected in the General tab.

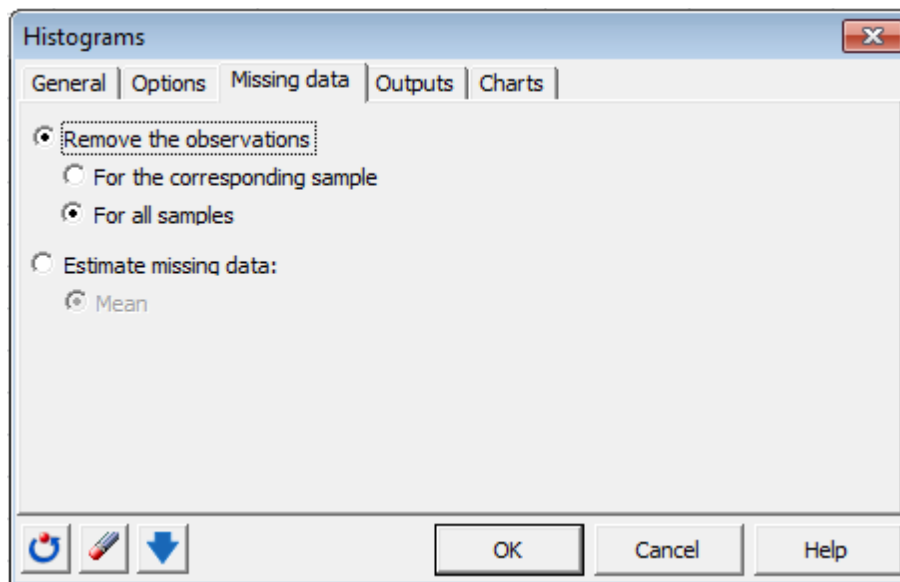
OPTIONS TAB

Use the Options tab to define the intervals for the histogram. These intervals may be defined by **Number:**, **Range:**, or **User defined:**. Activate the *Minimum* option to specify the lower value of the first interval.



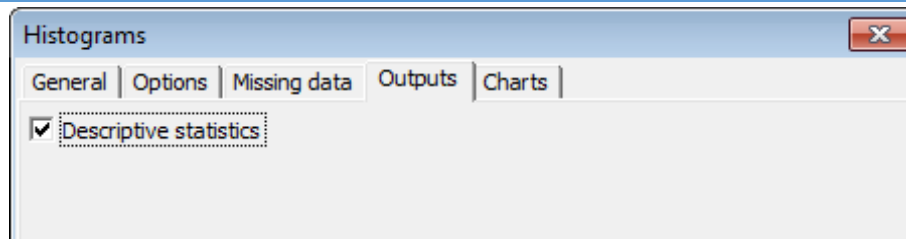
MISSING DATA TAB

The Missing data tab allows you to remove missing data or estimate missing data using the mean of the sample.



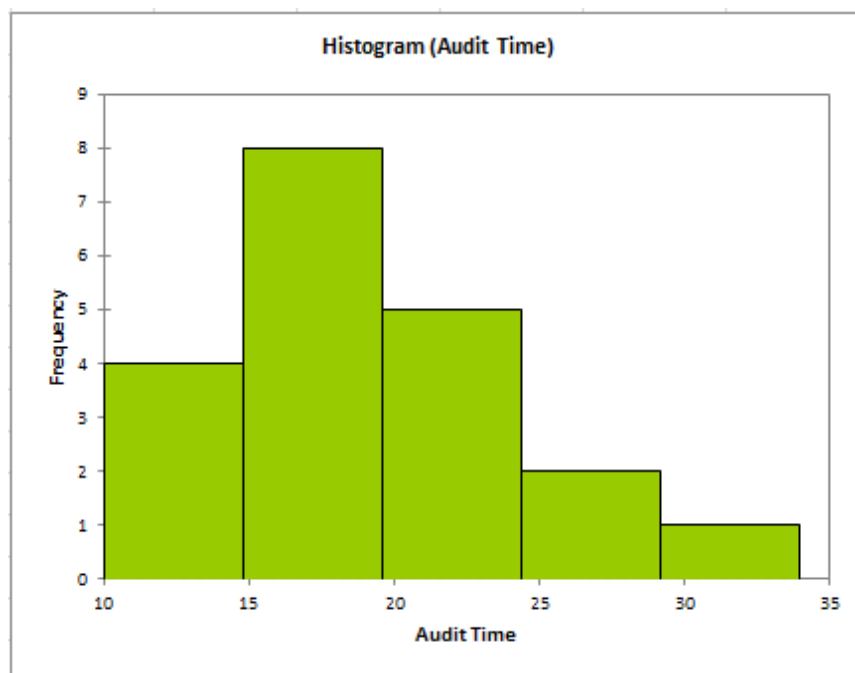
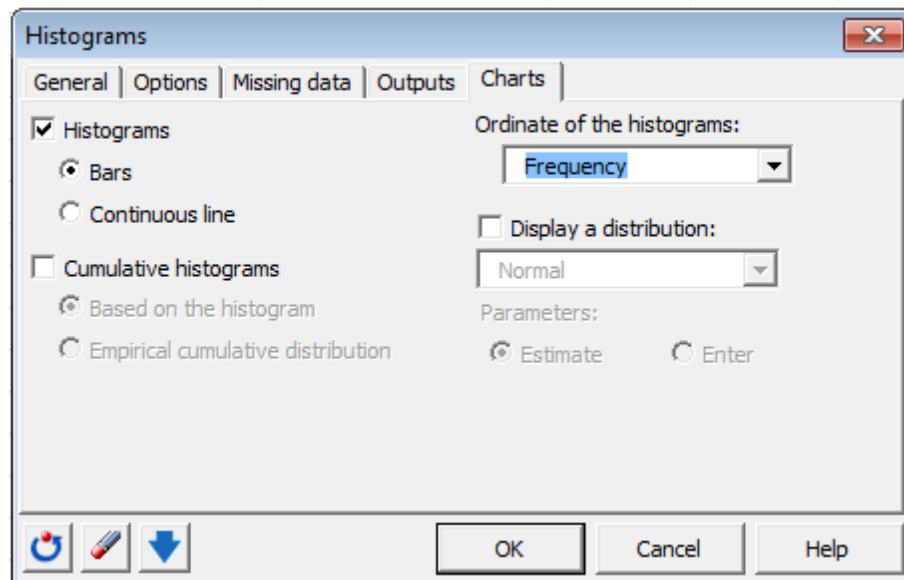
OUTPUTS TAB

The Outputs tab allows you to display the descriptive statistics of the samples (default).



CHARTS TAB

The Charts tab allows you to activate the display of the histogram. The histogram may be displayed as a bar or continuous line. The cumulative histograms option provides a display of cumulated histograms of the sample(s). The **Ordinate of the histograms:** provides the quantity to be used in the histogram as density, frequency, or relative frequency.

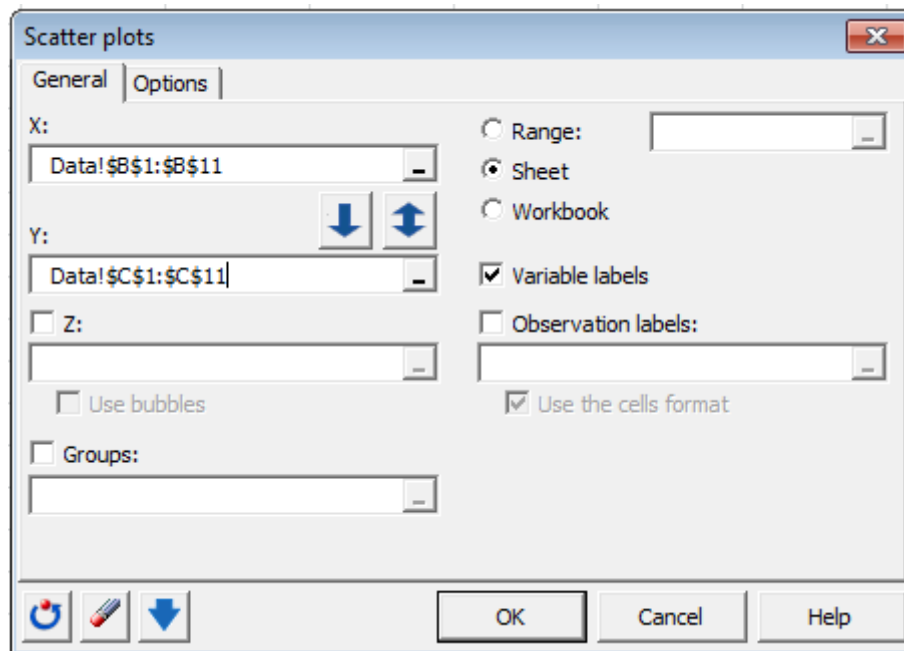


SCATTERPLOTS

Use this tool (found in the Visualizing Data tab) to create 2- or 3–dimensional plots (represented by the size of the point).

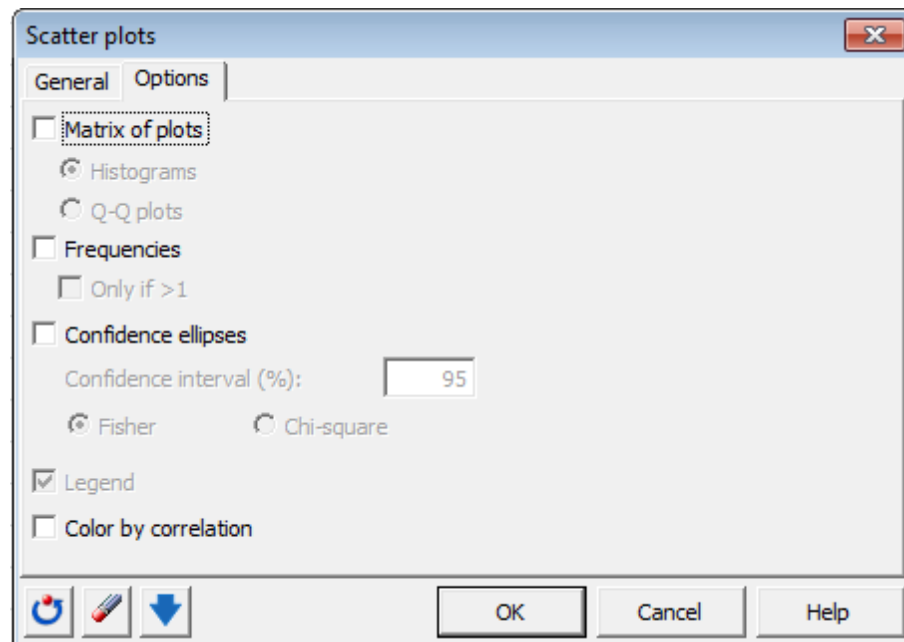
GENERAL TAB

In the General tab, specify the data along the X-axis in the **X:** field and the data along the Y-axis in the **Y:** field. The *Variable labels* checkbox frequencies for each point on the chart, and a display of confidence ellipses.



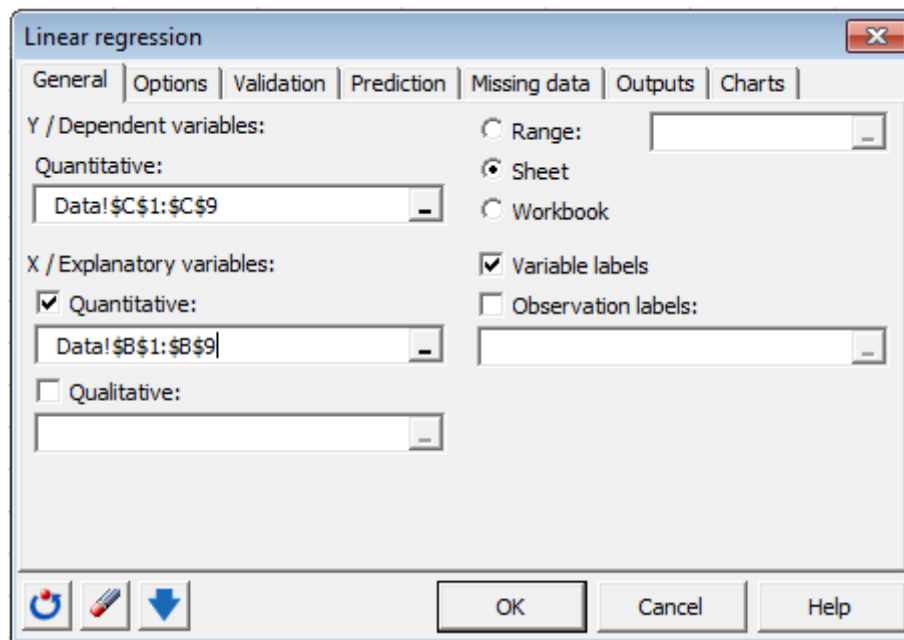
OPTIONS TAB

The Options tab displays by default a legend for the histogram. Additional options provide for a *Matrix of plots*, display of *Frequencies* for each point on the chart, and a display of *Confidence ellipses*.



LINEAR REGRESSION

Linear regression is the most frequently used statistical method. The principle of linear regression is to model a quantitative dependent variable Y through a linear combination of p quantitative explanatory variables, $X_1, X_2, \dots, X_p$. While a distinction is usually made between simple regression (with only one explanatory variable) and multiple regression (several explanatory variables), the overall concept and calculation methods are identical.

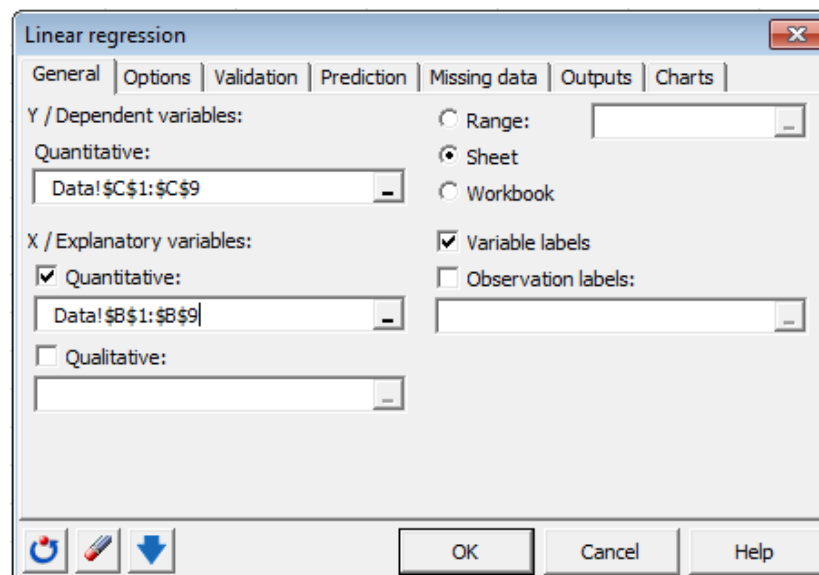


The screenshot shows the 'Linear regression' dialog box with the 'General' tab selected. The 'Y / Dependent variables:' section has 'Quantitative:' checked and 'Data!\$C\$1:\$C\$9' entered. The 'X / Explanatory variables:' section has 'Quantitative:' checked and 'Data!\$B\$1:\$B\$9' entered. The 'Variable labels' checkbox is checked, and 'Observation labels:' is unchecked. The 'Range:' radio button is unselected, and 'Sheet' is selected. The 'Workbook' radio button is unselected. The 'Qualitative:' checkbox is unchecked. The 'OK', 'Cancel', and 'Help' buttons are at the bottom right.

Use this tool (found in the Modeling Data tab) to create a simple or multiple linear regression model for explanation or prediction.

GENERAL TAB

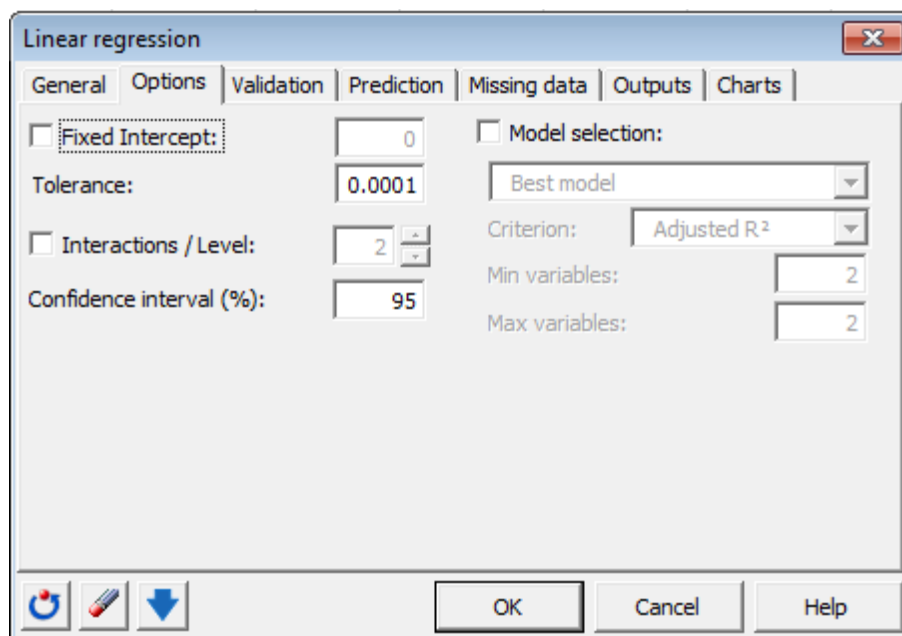
In the **Y / Dependent variables:** box select the response variable(s) you want to model. If several variables have been selected, XLSTAT carries out calculations for each of the variables separately. In the **X / Explanatory variables:** box select the quantitative explanatory variables in the Excel worksheet. The data selected must be of type numeric. Activate the **Qualitative:** data option and select qualitative variables to perform an ANCOVA analysis. Activate the *Variable labels* option if the first row of the data selections (dependent and explanatory variables, weights, observations labels) includes a header.



This is a duplicate of the screenshot above, showing the 'Linear regression' dialog box with the 'General' tab selected. The 'Y / Dependent variables:' section has 'Quantitative:' checked and 'Data!\$C\$1:\$C\$9' entered. The 'X / Explanatory variables:' section has 'Quantitative:' checked and 'Data!\$B\$1:\$B\$9' entered. The 'Variable labels' checkbox is checked, and 'Observation labels:' is unchecked. The 'Range:' radio button is unselected, and 'Sheet' is selected. The 'Workbook' radio button is unselected. The 'Qualitative:' checkbox is unchecked. The 'OK', 'Cancel', and 'Help' buttons are at the bottom right.

OPTIONS TAB

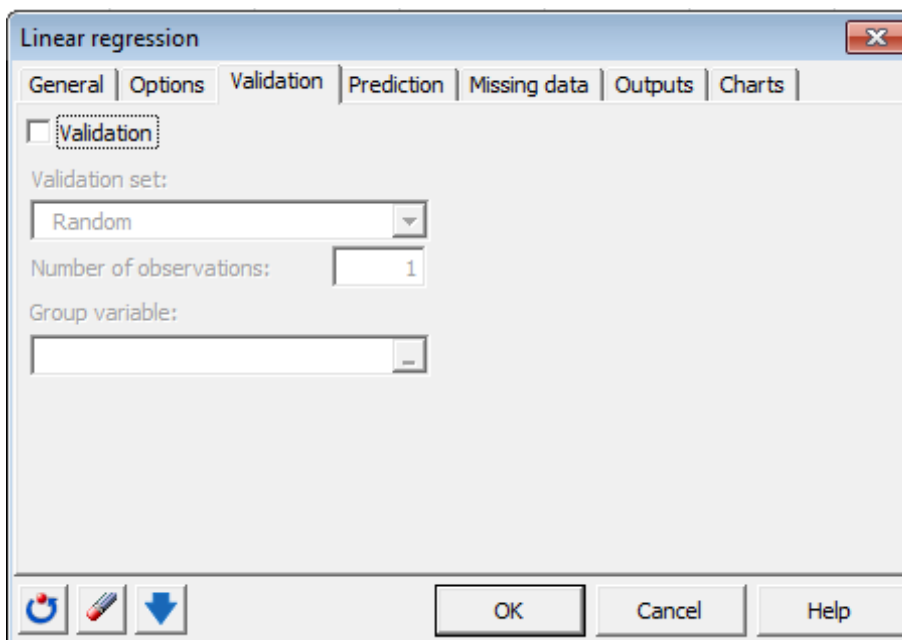
Activate the **Fixed Intercept**: option to set the regression constant to a fixed value (default 0). The **Tolerance**: option (default 0.0001) prevents the OLS regression algorithm from considering variables which are constant or too correlated with other variables in the model. The **Interactions/Level**: option allows you to include interactions in the model. Select a value between 1 and 4. Use the **Confidence interval (%)**: option to enter the percentage range of the confidence interval (default .95). Activate the **Model selection**: option if you want to indicate one of four available selection methods: best model regression, stepwise regression, forward, and backward regression.



The screenshot shows the 'Linear regression' dialog box with the 'Options' tab selected. The 'Fixed Intercept' checkbox is checked, with a value of 0. The 'Tolerance' is set to 0.0001. The 'Interactions / Level' checkbox is unchecked, with a value of 2. The 'Confidence interval (%)' is set to 95. The 'Model selection' checkbox is unchecked, with a dropdown menu set to 'Best model'. The 'Criterion' dropdown is set to 'Adjusted R²'. The 'Min variables' and 'Max variables' are both set to 2. The dialog box has a standard Windows interface with a title bar, tabs, and buttons at the bottom.

VALIDATION TAB

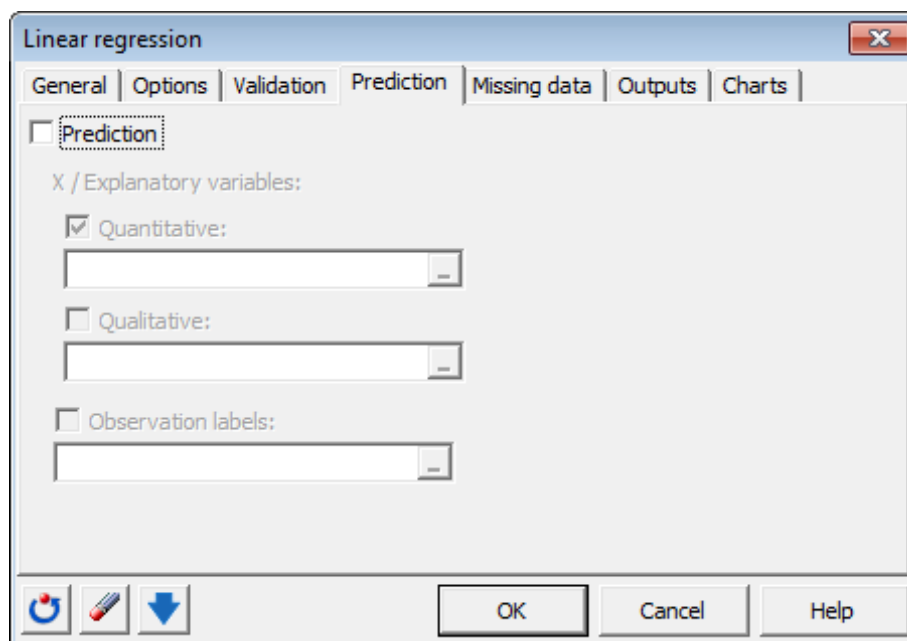
In the Validation tab, activate the Validation option if you want to use a sub-sample of data to validate the model. Define the method for obtaining the validation set as random, N last rows, N first rows, or Group variable.



The screenshot shows the 'Linear regression' dialog box with the 'Validation' tab selected. The 'Validation' checkbox is checked. The 'Validation set' dropdown is set to 'Random'. The 'Number of observations' is set to 1. The 'Group variable' dropdown is empty. The dialog box has a standard Windows interface with a title bar, tabs, and buttons at the bottom.

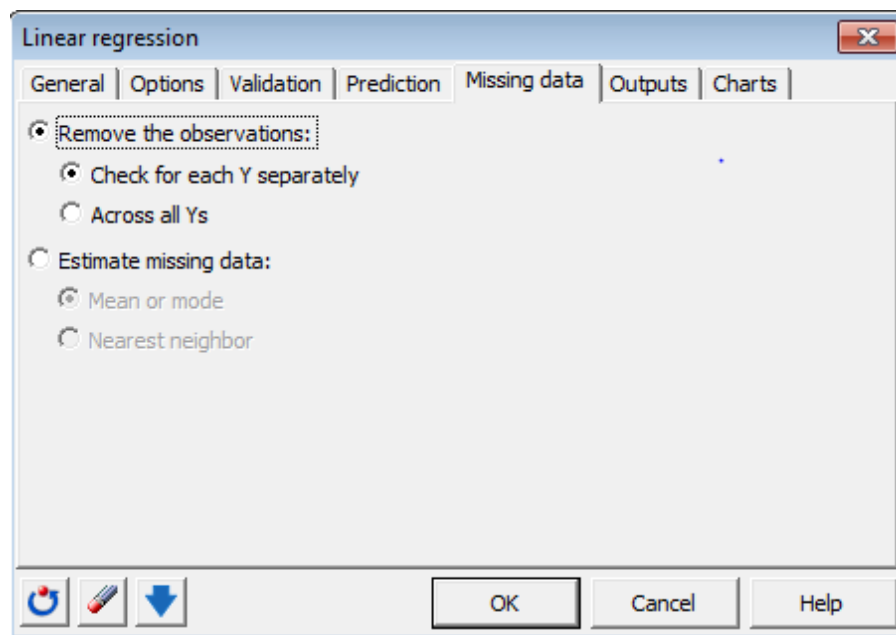
PREDICTION TAB

Activate the *Prediction* option in the Prediction tab to select data to use in prediction model. The data for Quantitative or Qualitative variables must not include variable labels.



MISSING DATA TAB

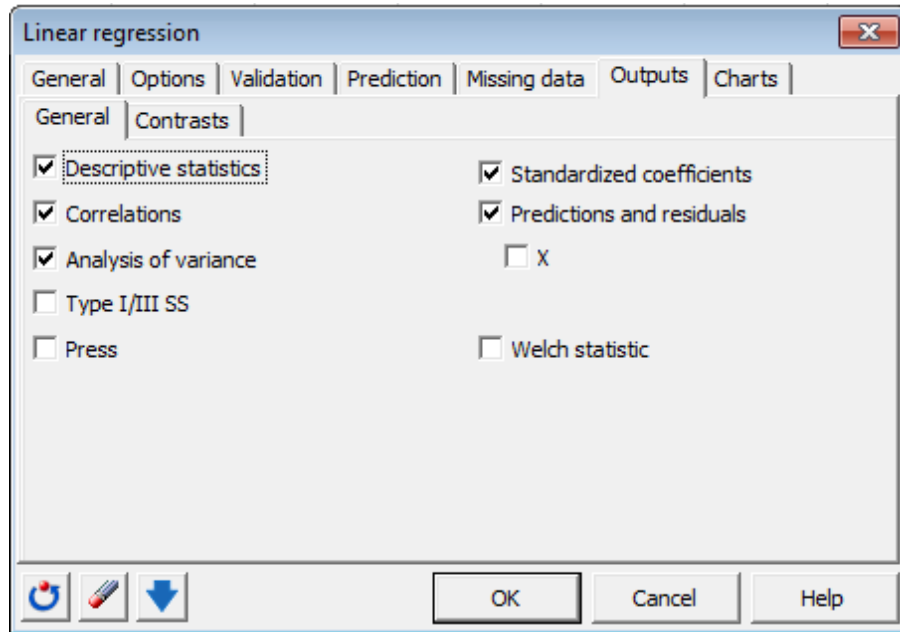
Activate the **Remove the observations:** option to remove the observations with missing data. The observations may be removed for each Y separately or across all Ys. Activate the **Estimate missing data:** option to estimate missing data before starting computations. The missing data may be estimated as the Mean or mode or Nearest neighbor.



OUTPUTS TAB

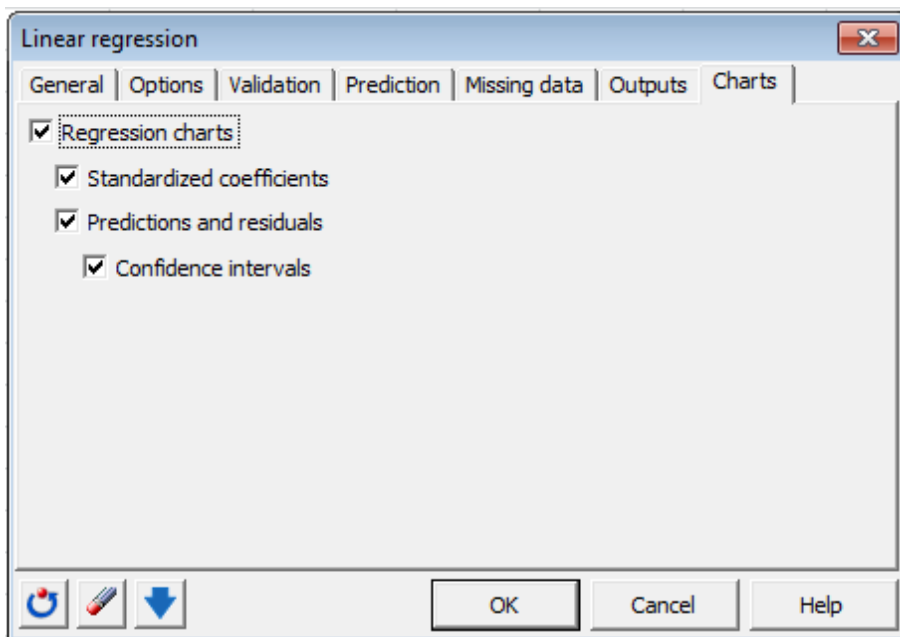
In the General sub-tab of the Outputs tab, defaults output is selected for *Descriptive statistics, Correlations, Analysis of variance, Standardized coefficients, Predictions and residuals*. In the Contrasts sub-tab of the Outputs tab, activate the Compute contrasts

option to compute contrasts, then select the contrasts table. There must be one column per contrast and one row for each coefficient of the model.



CHARTS TAB

In the Charts tab, *Regression charts*, *Standardized coefficients*, *Predictions and residuals*, and *Confidence intervals* are displayed by default. De-select any option to remove its display.



ANALYSIS OF VARIANCE (ANOVA)

Use this tool to carry out ANOVA (ANalysis Of VAriance) of one or more balanced or unbalanced factors.

ANOVA uses the same conceptual framework as linear regression. The presented dialog box options will be much the same. The main difference is that the explanatory variables are qualitative. Thus, required data organization for ANOVA in XLSTAT differs from data normally available for datasets provided with most texts. Consider the following data from National Computer Products, Inc. A

quality awareness examination was given to employees from three plants. Managers want to test the hypothesis that the mean examination score is the same for all three plants. You might receive (or expect to receive) the data in the following format:

Atlanta	Dallas	Seattle
85	71	59
75	75	64
82	73	62
76	74	69
71	69	75
85	82	67

XLSTAT expects that the data is organized as found in the Regression format where the quantitative data are in a single column. This data is delineated by city by qualitative data in an adjacent column as shown here:

E	F
Atlanta	85
Atlanta	75
Atlanta	82
Atlanta	76
Atlanta	71
Atlanta	85
Dallas	71
Dallas	75
Dallas	73
Dallas	74
Dallas	69
Dallas	82

GENERAL TAB

In the General tab, select the **Quantitative**: data you want to model as the Y/Dependent variables (data from column F above). In the X/Explanatory variables section choose **Qualitative**: explanatory variables (the factors from column E in the above example). Note that the data as presented does not include Variable labels and the output will be produced in a new worksheet.

ANOVA

General | Options | Validation | Prediction | Missing data | Outputs | Charts

Y / Dependent variables:

Quantitative: []

Range: []

Sheet [x]

Workbook []

X / Explanatory variables:

Quantitative: []

Qualitative: [x] []

Variable labels []

Observation labels: []

OK Cancel Help

OPTIONS TAB

Activate the **Fixed Intercept**: option to set the regression constant to a fixed value (default 0). The **Tolerance**: option (default 0.0001) prevents the OLS regression algorithm from considering variables which are constant or too correlated with other variables in the model. The **Interactions/Level**: option allows you to include interactions in the model. Select a value between 1 and 4. Use the **Confidence interval (%)**: option to enter the percentage range of the confidence interval (default .95).

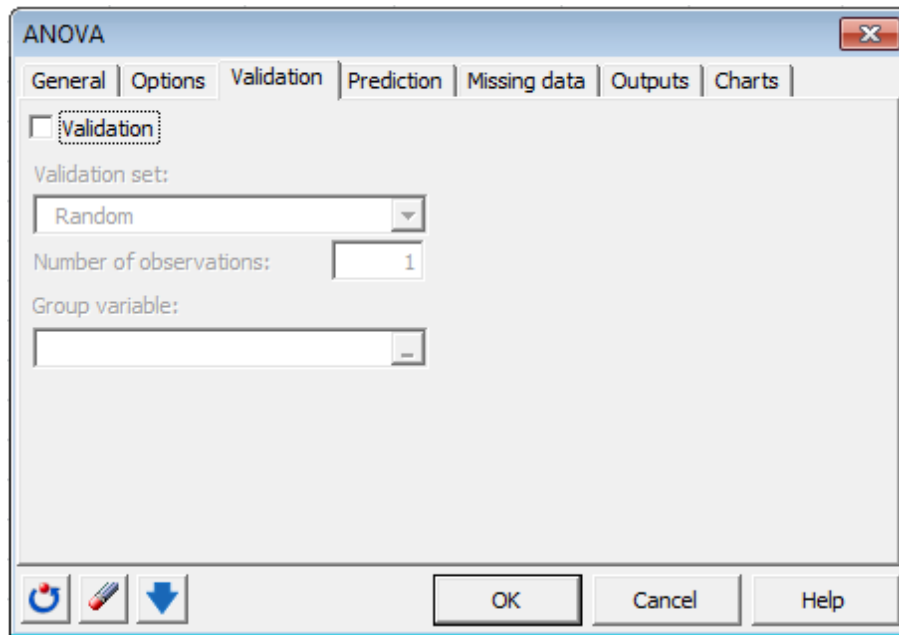
A number of **Constraints**: are available as $a_1 = 0$, $a_n = 0$ (default), $\sum (a_i) = 0$, and $\sum (n_i \cdot a_i) = 0$. These options are fully described in the Description section of the Help system in XLSTAT.

Activate the *Random effects* option to include random factors in the model displayed as an expected mean squares table.

The screenshot shows the 'ANOVA' dialog box with the 'Options' tab selected. The 'Fixed Intercept' checkbox is unchecked, with a value of 0. The 'Tolerance' is set to 0.0001. The 'Interactions / Level' checkbox is unchecked, with a value of 2. The 'Nested effects' checkbox is unchecked. The 'Confidence interval (%)' is set to 95. The 'Constraints' dropdown menu is set to 'an=0'. The 'Model selection' checkbox is unchecked, with a dropdown menu set to 'Best model'. The 'Criterion' dropdown menu is set to 'Adjusted R²'. The 'Min variables' and 'Max variables' are both set to 2. The 'Random effects' checkbox is unchecked. At the bottom, there are icons for a refresh, a pencil, and a download, followed by 'OK', 'Cancel', and 'Help' buttons.

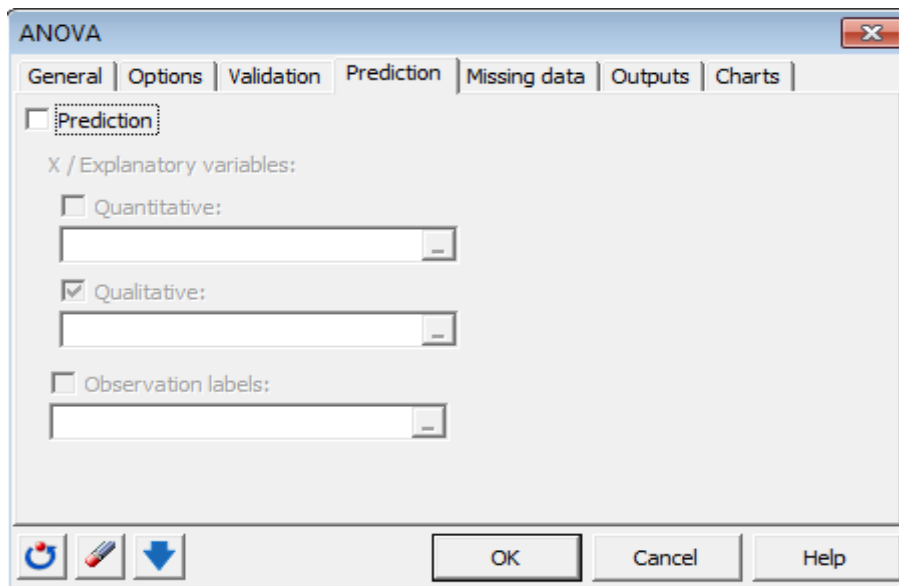
VALIDATION TAB

Activate this option to use a sub-sample of the data to validate the model. Define how to obtain the observations as random, N last rows, N first rows, or a group variable which is a binary with the 1s option identifying the observations to use for the validation.



PREDICTION TAB

Activate the *Prediction* option in the Prediction tab to select data to use in prediction model. The data for **Quantitative:** or **Qualitative:** variables must not include variable labels.



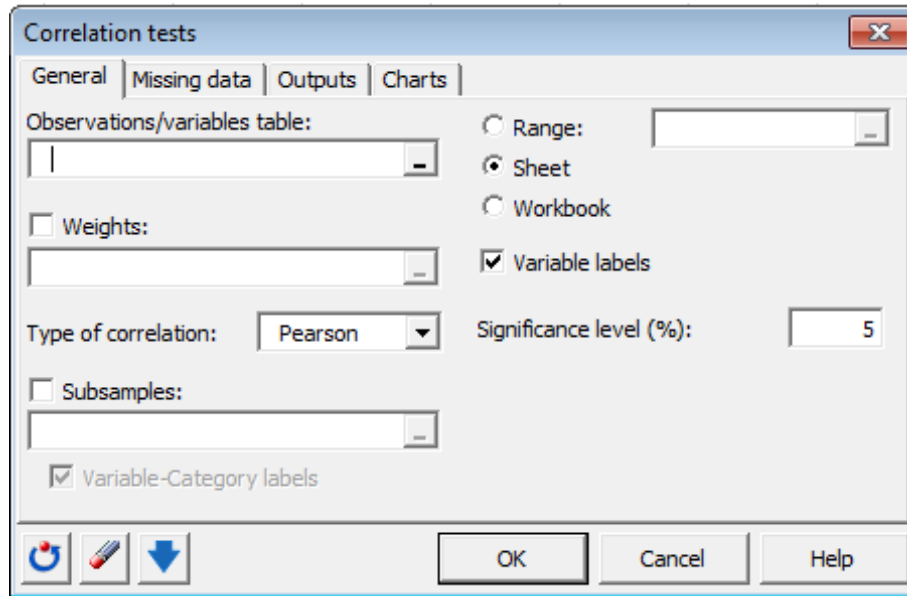
CORRELATION TESTS

Use this tool (in the Correlation/Association tests tab) to compute the correlation coefficients of Pearson, Spearman or Kendall, between two or more variables, and to determine if the correlations are significant or not. Several visualizations of the correlation matrices are proposed. The Pearson correlation coefficient corresponds to the classical linear correlation coefficient for continuous data. Its value ranges from -1 to 1, and it measures the degree of linear correlation between two variables.

GENERAL TAB

Select a table in Excel of N observations (rows) described by P variables (columns) in the **Observations/variables table:** option. Activate the **Weights:** option as a number greater than or equal to 0 if the observations are weighted. Check the **Subsamples:** option

to select a column showing the names or indexes of the subsamples for each of the observations. All computations are then performed.



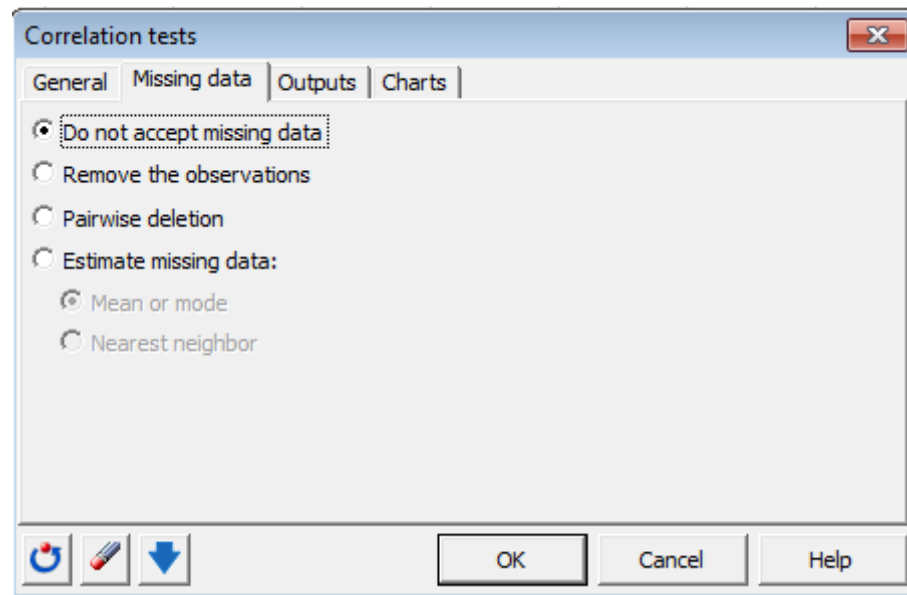
The 'Correlation tests' dialog box is shown with the 'General' tab selected. It contains the following settings:

- Observations/variables table:** A text box with a dropdown arrow.
- Weights:** A checkbox (unchecked) and a text box with a dropdown arrow.
- Type of correlation:** A dropdown menu set to 'Pearson'.
- Subsamples:** A checkbox (unchecked) and a text box with a dropdown arrow.
- Range:** A radio button (unchecked) and a text box with a dropdown arrow.
- Sheet:** A radio button (checked).
- Workbook:** A radio button (unchecked).
- Variable labels:** A checkbox (checked).
- Significance level (%):** A text box set to '5'.
- Variable-Category labels:** A checkbox (checked).

At the bottom, there are three icons (a circular arrow, a pencil, and a downward arrow) and three buttons: 'OK', 'Cancel', and 'Help'.

MISSING DATA TAB

Use the Missing data tab to determine how missing data should be treated in calculations: *Do not accept missing data*, *Remove the observations*, *Pairwise deletion* or *Estimate missing data* using the mean or mode or nearest neighbor.



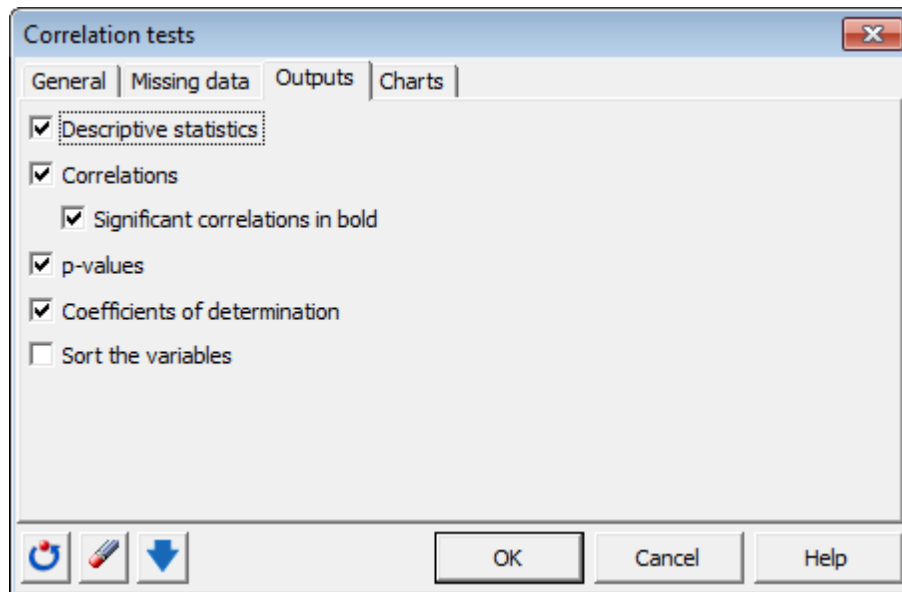
The 'Correlation tests' dialog box is shown with the 'Missing data' tab selected. It contains the following settings:

- Do not accept missing data:** A radio button (checked).
- Remove the observations:** A radio button (unchecked).
- Pairwise deletion:** A radio button (unchecked).
- Estimate missing data:** A radio button (unchecked) with two sub-options:
 - Mean or mode:** A radio button (checked).
 - Nearest neighbor:** A radio button (unchecked).

At the bottom, there are three icons (a circular arrow, a pencil, and a downward arrow) and three buttons: 'OK', 'Cancel', and 'Help'.

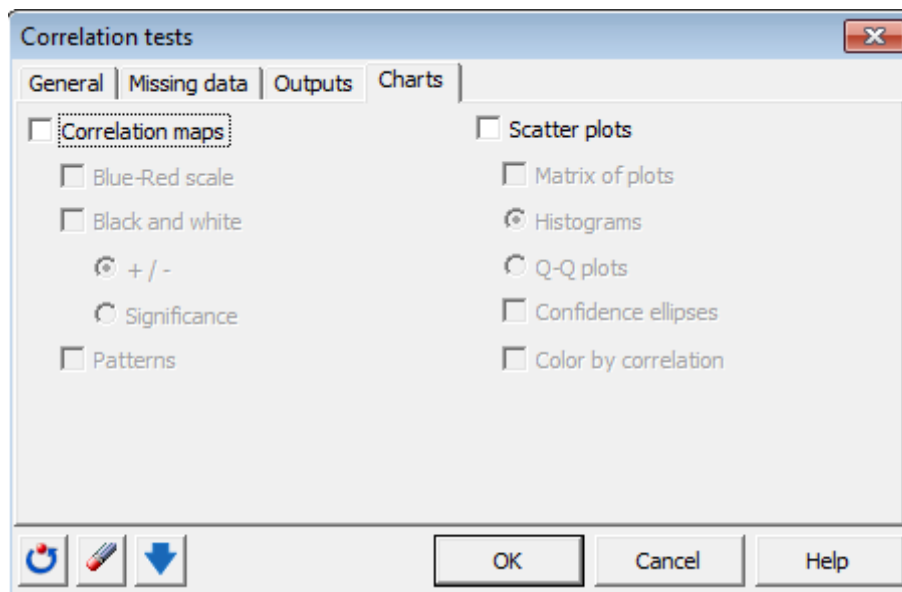
OUTPUTS TAB

In the Output tab, default settings include descriptive statistics, correlations, p-values, and coefficients of determination. Choose the *Sort the variables* option to provide the data sorted alphabetically by variable name.



CHARTS TAB

Use the charts tab to visualize the correlation matrix as a set of *Correlation maps* or *Scatter plots*. Correlation maps may be provided in blue-red, black and white, or patterns scales to represent positive and negative correlations. Present scatter plots as histogram or Q-Q matrix plots or confidence ellipses, for a x% confidence interval for a bivariate normal distribution. Additionally colors by correlation may be added.



PARAMETRIC TESTS

Use this tool (in the Parametric tests tab) to conduct one- and two-sample t and z tests, one-and two-sample variance tests, z-tests for one and two proportions, and the multinomial goodness of fit test. Tests are said to be parametric since their use requires the assumption that the samples are distributed normally. Additionally, it is also assumed that observations are independent and identically distributed.

ONE-SAMPLE T-TEST AND Z-TEST

Use this tool to compare the mean of a normally-distributed sample with a given value. Two parametric tests are possible. **Student's t test** if the true variance of the population from which the sample has been extracted is not known; the variance of sample s^2 is used as the variance estimator. Use the **z test** if the true variance s^2 of the population is known. Three types of test are possible depending on the alternative hypothesis chosen:

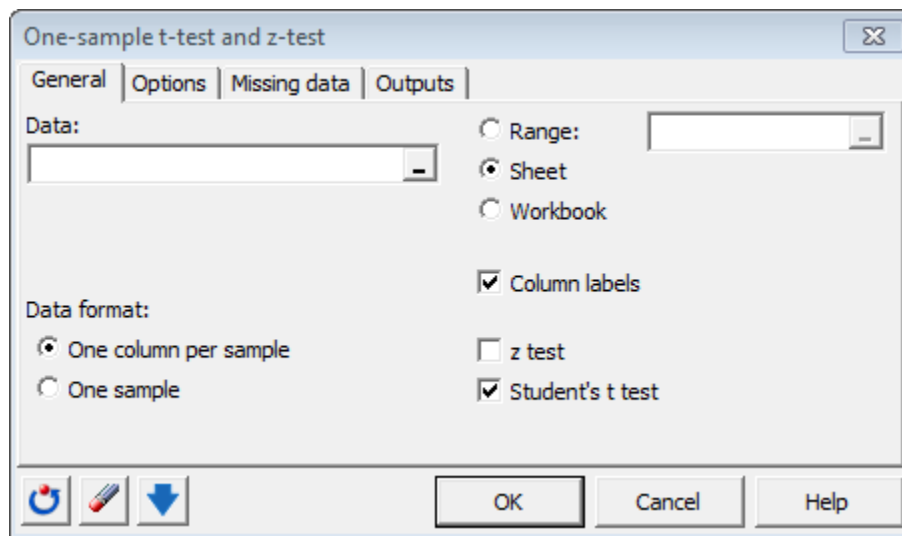
Two-tailed test $H_a: \mu \neq \mu_0$

Left one-tailed test $H_a: \mu < \mu_0$

Right one-tailed test $H_a: \mu > \mu_0$

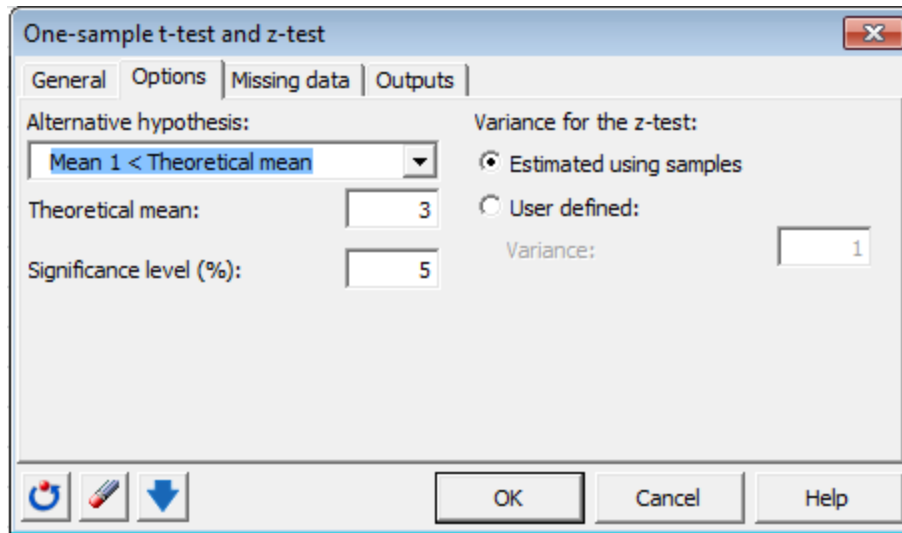
GENERAL TAB

Select the data in the Excel worksheet. Choose the **Data format:** as *One column/row per sample* which considers each column or row as a sample and allows for testing hypotheses on several samples at the same time OR *One sample*: to consider all values as a single sample. Activate the *Column/row labels* option if the selected data contain labels. Activate the *z Test* or *Student's t Test* option depending on the type of test to be conducted.



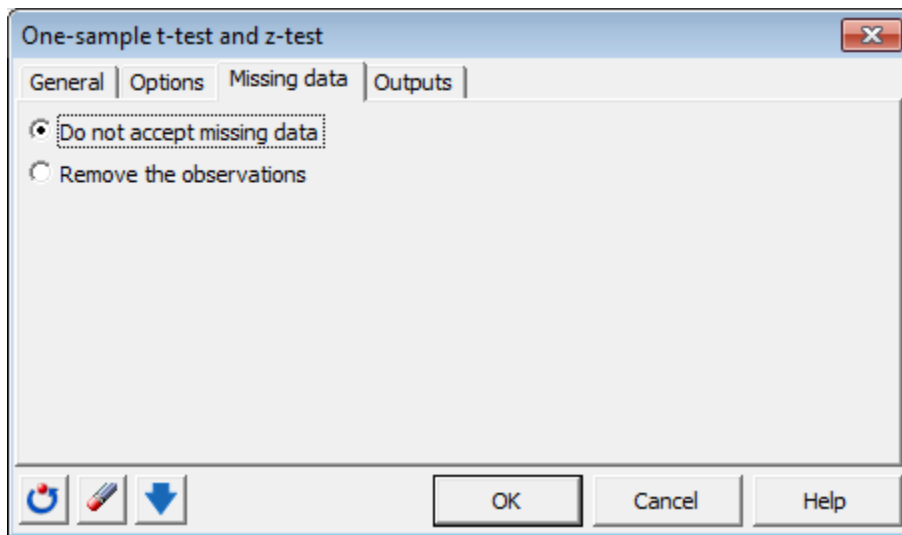
OPTIONS TAB

Choose the **Alternative hypothesis:** to be used for the test. Enter the value of the **Theoretical mean:** with which the mean of the sample is to be compared, and the **Significance level (%):** for the tests (default value: 5%). If the z test option has been selected, indicate whether the **Variance for the z-test:** should be *Estimated using samples* or *User defined*: which requires a value to be entered for the variance.



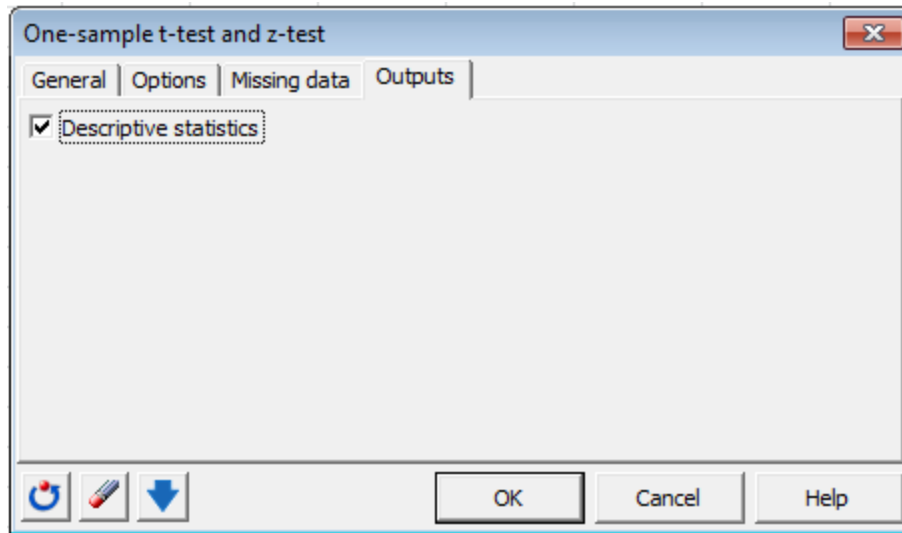
MISSING DATA TAB

Activate the desired option: *Do not accept missing data* OR *Remove the observations*. If the missing data option is activated, XLSTAT will discontinue calculations if missing values are detected. The remove observations option will remove observations with missing data and continue processing.



OUTPUTS TAB

The Outputs tab allows you to display the descriptive statistics of the samples (default).



TWO-SAMPLE T-TEST AND Z-TEST

Use this tool to compare the means of two normally distributed independent or paired samples. A distinction is made between independent samples (for example a comparison of annual sales by shop between two regions for a chain of supermarkets), or paired samples (for example if comparing the annual sales within the same region over two years). When comparing the means of 2 independent samples, use Student's t test if the true variance of the populations from which the samples are extracted is not known; the z test if the true variance s^2 of the population is known. When the Student's t Test is used a decision must be made about the equality/inequality of variances. When comparing the means of two paired samples, sample sizes of the paired samples must be the same size. Where values are missing from certain observations, either the observation is removed from both samples or the missing values are estimated.

Three types of test are possible depending on the alternative hypothesis chosen

Two-tailed test $H_a: u_1 - u_2 \neq D$

Left one-tailed test $H_a: u_1 - u_2 < D$

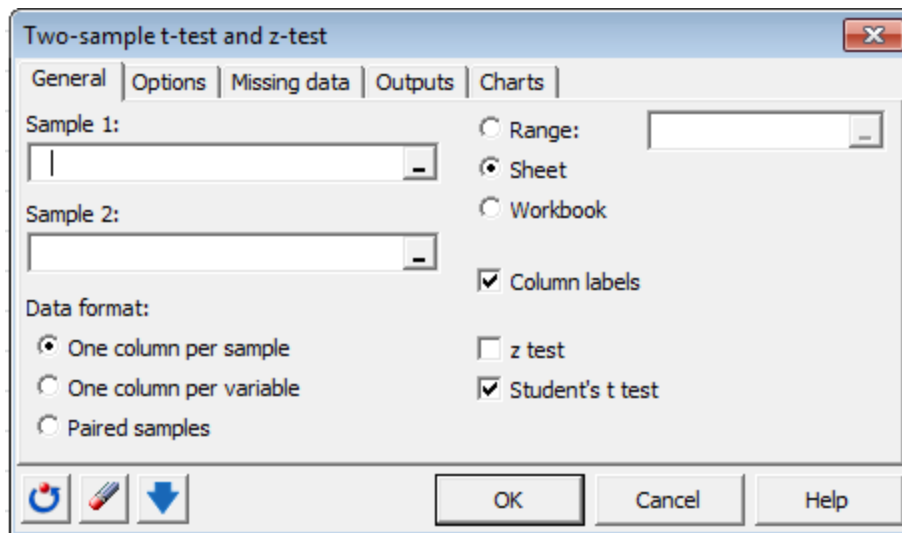
Right one-tailed test $H_a: u_1 - u_2 > D$

GENERAL TAB

Sample 1: If the format of the selected data is "one column per variable", select the data for the various samples in the Excel worksheet. If the format of the selected data is "one column per sample" or "paired samples", select a column of data corresponding to the first sample.

Sample 2: If the format of the selected data is "one column per variable", select the data identifying the two samples to which the selected data values correspond. If the format of the selected data is "one column per sample" or "paired samples", select a column of data corresponding to the second sample.

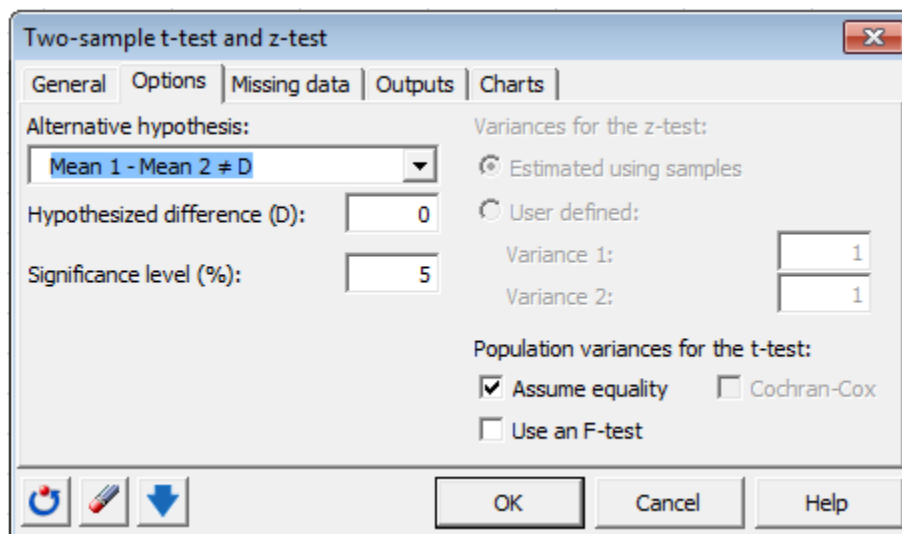
Choose the **Data format:** as *One column/row per sample*, *One column/row per variable*, or *Paired samples*. Activate the *Column/row labels* option if the first row/column of the selected data contain labels. Activate the *z-test* or *Student's t test* to indicate the type of test to execute.



OPTIONS TAB

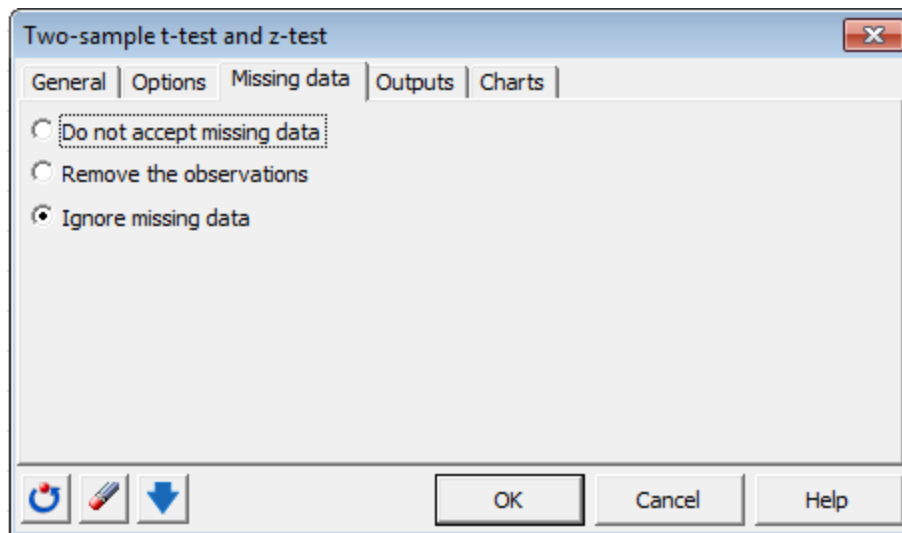
Choose the **Alternative hypothesis:** to be used for the test. Enter the value of the supposed difference between the samples as the **Hypothesized difference (D):**. Enter the significance level for the tests (default value: 5%) as the **Significance level (%):**. A **Weights:** option is available if the data format is "One column/row per variable" or if the data are paired. Check this option if the observations are weighted.

Where a z test has been requested, the value of the known variance of the populations, or, for a test on paired samples, the variance of the difference must be entered. Choose the *Estimated using samples* option to estimate the variance of the population from the sample data (leads to a t test). Choose the *User defined:* option to enter the values of the known variances of the populations. For t-test samples variances, activate *Assume equality* to consider sample variances equal, *Cochran-Cox* to calculate the p-value using the Cochran and Cox method where the variances are assumed to be unequal, and *Use an F test* to use Fisher's F test to determine whether the variances of both samples can be considered to be equal or not.



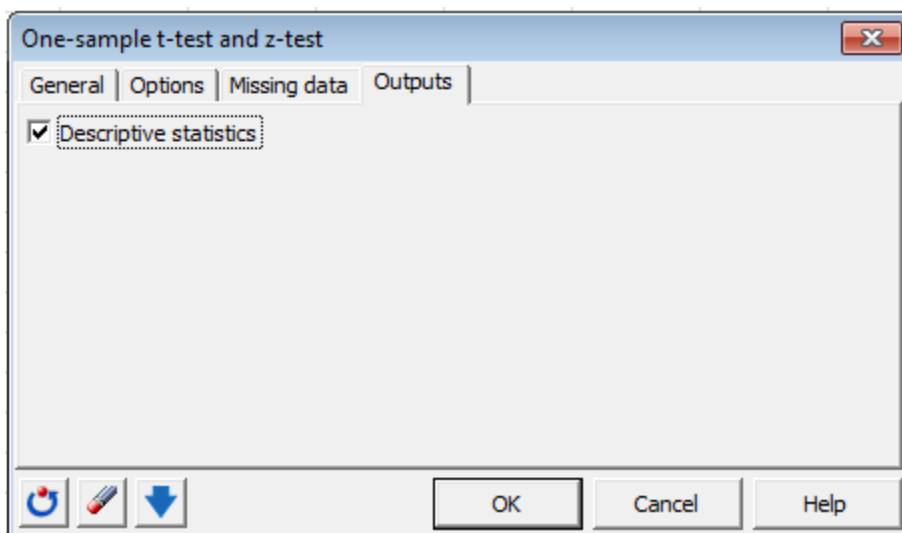
MISSING DATA

In the Missing data tab, activate the appropriate option to not accept missing data, remove the observations, or ignore missing data (default).



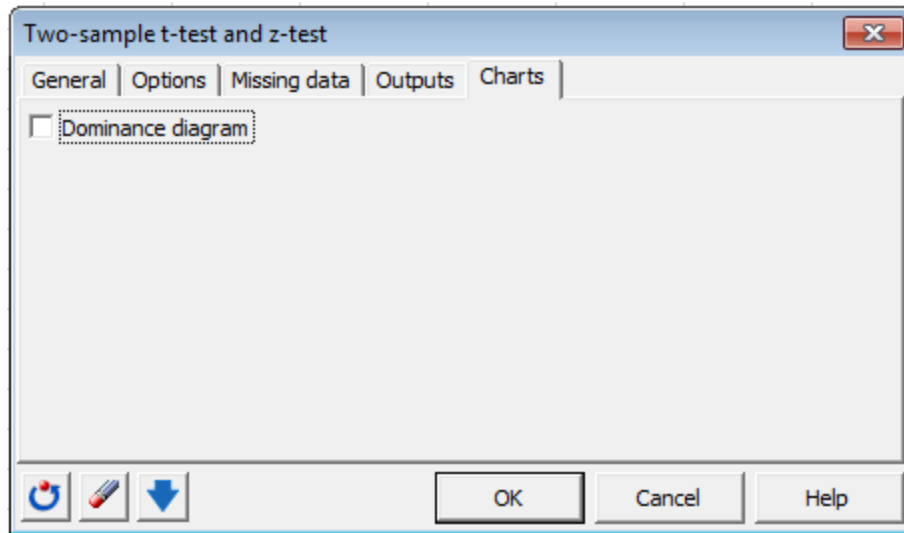
OUTPUTS TAB

The Outputs tab allows you to display the descriptive statistics of the samples (default).



CHARTS TAB

Activate the **Dominance diagram**: option to display a dominance diagram in order to make a visual comparison of the samples.



ONE-SAMPLE VARIANCE TEST

Use this tool to compare the variance of a normally-distributed sample with a given value. The sample variance, s^2 , follows a scale chi-squared distribution with $n-1$ degrees of freedom. Three types of test are possible depending on the alternative hypothesis chosen:

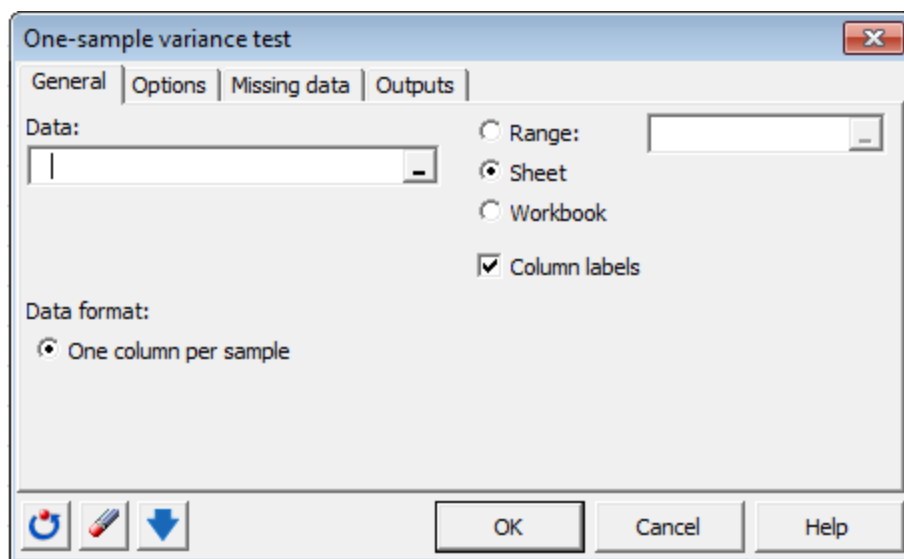
For the two-tailed test, the alternative hypothesis is $H_a: \sigma^2 \neq \sigma_0^2$

In the left-one-tailed test, the alternative hypothesis is $H_a: \sigma^2 < \sigma_0^2$

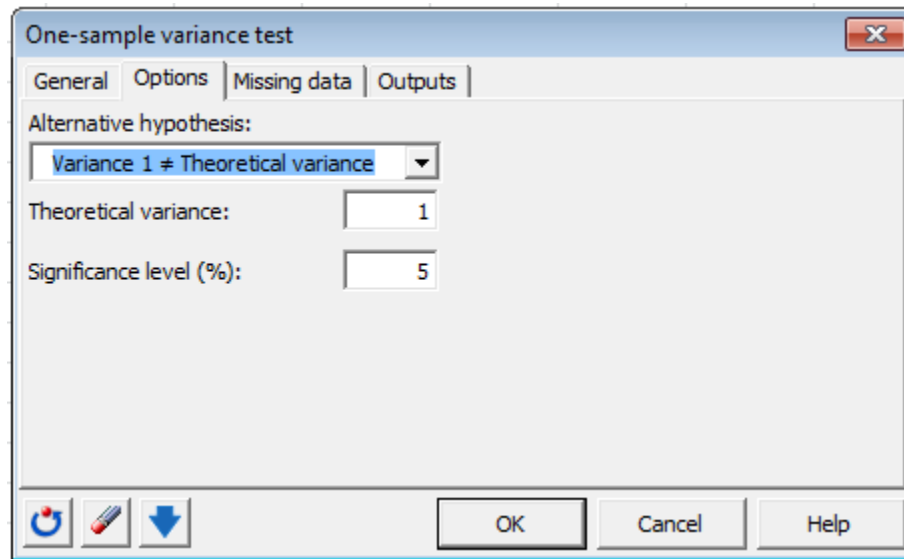
In the right one-tailed test, the alternative hypothesis is $H_a: \sigma^2 > \sigma_0^2$

GENERAL TAB

Select the **Data:** in the Excel worksheet. Activate the **Data format:** option to select one column per sample when multiple samples are available in the dataset. Activate the *Column/row labels* option if the selected data contain labels.



OPTIONS TAB



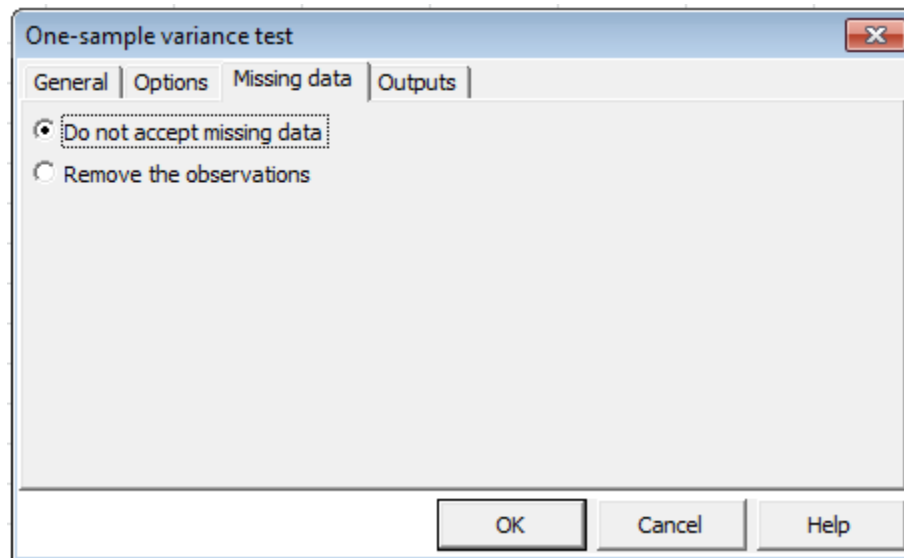
Choose the **Alternative hypothesis**: to be used for the test.

Theoretical variance: Enter the value of the theoretical mean with which the mean of the sample is to be compared.

Significance level (%): Enter the significance level for the tests (default value: 5%).

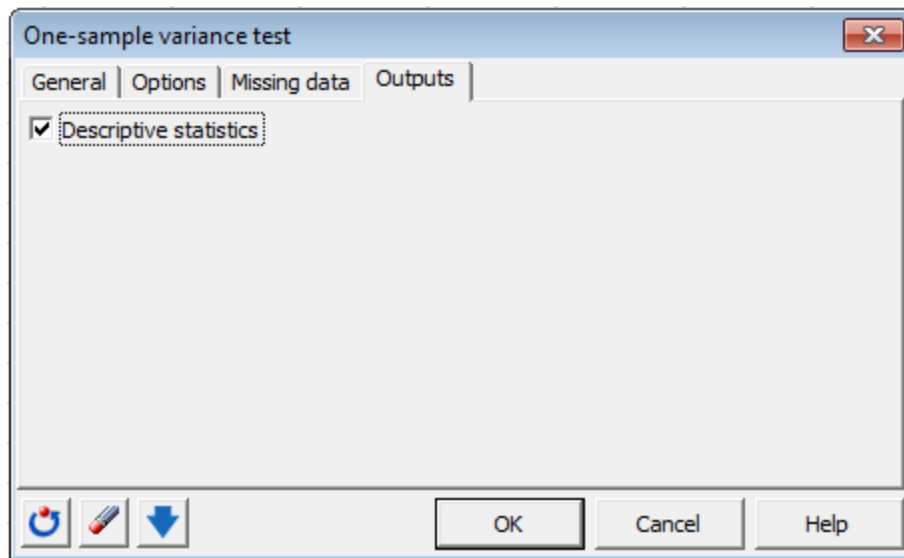
MISSING DATA

Activate the appropriate option to not accept missing data or remove the observations with missing data.



OUTPUTS TAB

The Outputs tab allows you to display the descriptive statistics of the samples (default).



TWO-SAMPLE COMPARISON OF VARIANCE

Use this tool to compare the variances of two samples. Take a sample S_1 comprising n_1 observations with variance s_1^2 . Take a second sample S_2 comprising n_2 observations with variance s_2^2 . XLSTAT offers three tests for comparing the variances of the two samples.

Fisher's F test

The test statistic F is given by: $F = s_1^2 / s_2^2$. This statistic follows a Fisher distribution with (n_1-1) and (n_2-1) degrees of freedom if both samples follow a normal distribution. Three types of test are possible depending on the alternative hypothesis chosen:

For the two-tailed test, the alternative hypothesis is $H_a: \sigma_1^2 \neq \sigma_2^2$

In the left-one-tailed test, the alternative hypothesis is $H_a: \sigma_1^2 < \sigma_2^2$

In the right one-tailed test, the alternative hypothesis is $H_a: \sigma_1^2 > \sigma_2^2$

Levene's test

Levene's test is a two-tailed test for which the null and alternative hypotheses are given by the following for the case where two variances are being compared:

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_a: \sigma_1^2 \neq \sigma_2^2$$

The statistic from the Levene test is more complex than that from the Fisher test. It follows a Fisher's F distribution with 1 and $n_1 + n_2 - 2$ degrees of freedom.

Bartlett's homogeneity of variances test

Bartlett's test is sensitive to the normality of the data. In other words, if the hypothesis of normality of the data seems fragile, it is better to use Levene's or Fisher's test. On the other hand, Bartlett's test is more powerful if the samples follow a normal distribution. Bartlett's statistic follows a Chi-square distribution with one degree of freedom.

GENERAL TAB

Sample 1: If the format of the selected data is "one column per variable", select the data for the various samples in the Excel worksheet. If the format of the selected data is "one column per sample" or "paired samples", select a column of data corresponding to the first sample.

Sample 2: If the format of the selected data is "one column per variable", select the data identifying the two samples to which the selected data values correspond. If the format of the selected data is "one column per sample" or "paired samples", select a column of data corresponding to the second sample.

Choose the **Data format:** option as *One column/row per sample* or *One column per variable*. Activate the *Column/row labels* option if the first column (or row) of the selected data contains labels.

Activate the *Fisher's F-test*, *Levene's test*, or *Bartlett's test* to indicate the test statistic to compute.

The screenshot shows the 'Two-sample comparison of variances' dialog box with the 'General' tab selected. The 'Sample 1:' and 'Sample 2:' fields are empty. The 'Data format:' section has 'One column per sample' selected. The 'Options' section has 'Fisher's F-test' checked, 'Levene's test' unchecked, and 'Bartlett's test' unchecked. The 'Mean' radio button is selected under the test options. The 'Column labels' checkbox is checked. The 'Range:', 'Sheet', and 'Workbook' options are not selected. The 'OK', 'Cancel', and 'Help' buttons are at the bottom right.

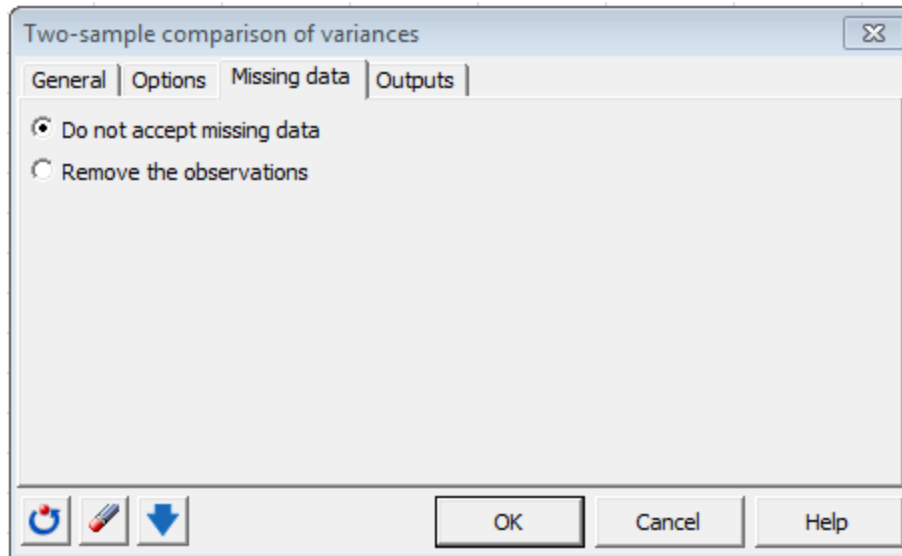
OPTIONS TAB

Choose the **Alternative hypothesis:** to be used for the test. Enter the value of the supposed ratio between the variances of the samples (typically 1) in the **Hypothesized ratio (R):** option. Enter the significance level for the tests (default value: 5%) as the **Significance level (%)**:

The screenshot shows the 'Two-sample comparison of variances' dialog box with the 'Options' tab selected. The 'Alternative hypothesis:' dropdown menu is set to 'Variance 1 / Variance 2 ≠ R'. The 'Hypothesized ratio (R):' field contains the value '1'. The 'Significance level (%)' field contains the value '5'. The 'OK', 'Cancel', and 'Help' buttons are at the bottom right.

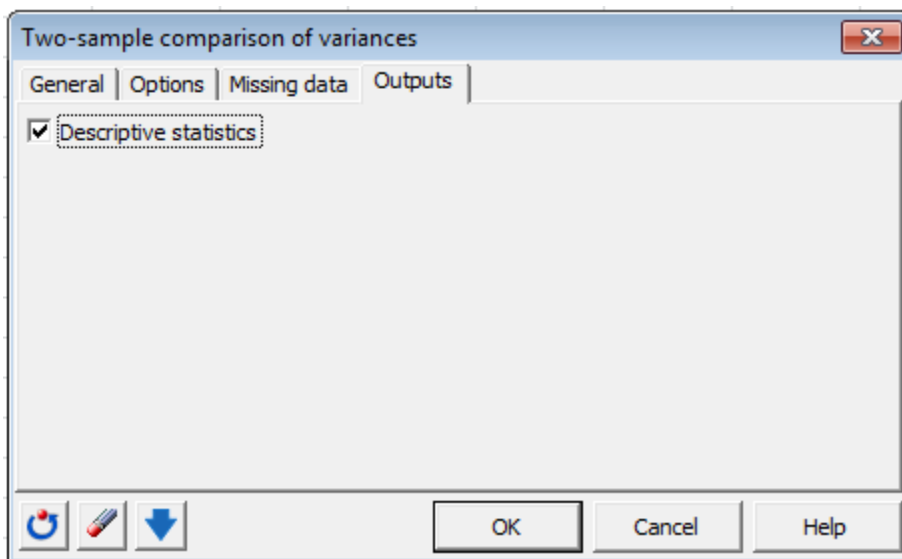
MISSING DATA TAB

Activate the appropriate option to not accept missing data or remove the observations with missing data.



OUTPUTS TAB

The Outputs tab allows you to display the descriptive statistics of the samples (default).



Z-TEST FOR ONE PROPORTION

Use this test to compare a proportion calculated from a sample with a given proportion. Let n be the number of observations verifying a certain property among a sample of size N . The proportion of the sample verifying the property is defined by $p = n / N$. Let p_0 be a known proportion with which we wish to compare p and D be the assumed difference ($p - p_0$). Three types of test are possible depending on the alternative hypothesis chosen:

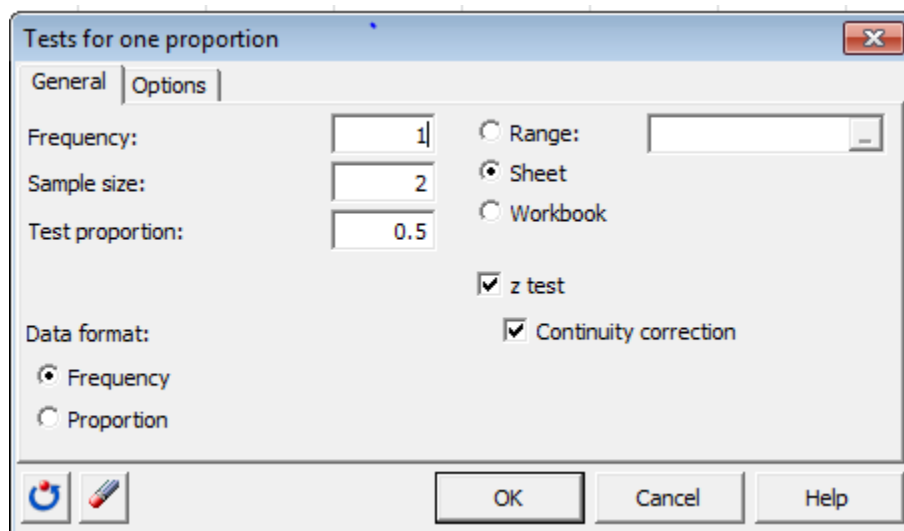
For the two-tailed test, the alternative hypothesis is $H_a: p - p_0 \neq D$

In the left-one-tailed test, the alternative hypothesis is $H_a: p - p_0 < D$

In the right one-tailed test, the alternative hypothesis is $p - p_0 > D$

GENERAL TAB

In the **Data format:** area activate *Frequency* if you would prefer to enter the value of the number of observations for which the property is observed, or activate *Proportion* to enter the proportion observed. Based on your option in the Data format: section, indicate the **Frequency:** as the number of observations or corresponding proportion. Enter the number of observations in the **Sample size:** option. Enter the value of the test proportion with which the proportion observed is to be compared in the **Test Proportion:** option. The *z test* and *Continuity correction* options are automatically selected. To deselect an option, click on the check box.



The screenshot shows a dialog box titled "Tests for one proportion" with a close button (X) in the top right corner. It has two tabs: "General" (selected) and "Options". In the "General" tab, there are three input fields: "Frequency:" with the value "1", "Sample size:" with the value "2", and "Test proportion:" with the value "0.5". To the right of these fields are three radio buttons: "Range:" (unselected), "Sheet" (selected), and "Workbook" (unselected). Below these are two checked checkboxes: "z test" and "Continuity correction". At the bottom left, under "Data format:", there are two radio buttons: "Frequency" (selected) and "Proportion" (unselected). At the bottom right are three buttons: "OK", "Cancel", and "Help". There are also two small icons (a circular arrow and a pencil) on the bottom left.

OPTIONS TAB

Choose the **Alternative hypothesis:** to be used for the test. Enter the value of the supposed difference between the proportions of the sample in the **Hypothesized ratio (D):** option. Enter the significance level for the tests (default value: 5%) as the **Significance level (%)**:. Choose the **Variance (confidence interval):** method to estimate the variance of the proportion. Activate *Sample* to compute the variance using the proportion obtained for the sample. Activate *Test proportion* to compute the variance using the test proportion and the size of the sample. Select the method used to compute the **Confidence interval:** as *Wald*, *Wilson score*, *Clopper-Pearson*, or *Agresti-Goull*.

Tests for one proportion

General Options

Alternative hypothesis:
 Proportion - Test proportion $\neq D$

Hypothesized difference (D): 0

Significance level (%): 5

Confidence interval:
☐ Wald
☒ Wilson score
☐ Clopper-Pearson
☐ Agresti-Coull

Variance (confidence interval):
☒ Sample
☐ Test proportion

OK Cancel Help

Z-TEST FOR TWO PROPORTIONS

Use this tool to compare two proportions calculated for two samples. Let n_1 be the number of observations verifying a certain property for sample S_1 of size N_1 , and n_2 the number of observations verifying the same property for sample S_2 of size N_2 . The proportion of sample S_1 verifying the property is defined by $p_1 = n_1 / N_1$, and the proportion for S_2 is defined by $p_2 = n_2 / N_2$. Let D be the assumed difference (exact, minimum or maximum) between the two proportions p_1 and p_2 . D is usually 0. Three types of test are possible depending on the alternative hypothesis chosen:

For the two-tailed test, the alternative hypothesis is $H_a: p_1 - p_2 \neq D$

In the left-one-tailed test, the alternative hypothesis is $H_a: p_1 - p_2 < D$

In the right one-tailed test, the alternative hypothesis is $H_a: p_1 - p_2 > D$

GENERAL TAB

In the **Data format:** area activate *Frequencies* if you would prefer to enter the value of the number of observations for which the property is observed, or activate *Proportions* to enter the proportion observed. Based on your option in the Data format: section, indicate the **Frequency 1:** as the number of observations or corresponding proportion for Sample 1. Enter the number of observations for Sample 1 in the **Sample size 1:** option. Indicate the **Frequency 2:** as the number of observations or corresponding proportion for Sample 2. Enter the number of observations for Sample 2 in the **Sample size 2:** option. Enter the value of the test proportion with which the proportion observed is to be compared in the **Test Proportion:** option. The *z test* option is automatically selected. Activate the **Monte Carlo method: option** to compute the p-value using Monte Carlo simulations. Enter the number of simulations to perform.

Tests for two proportions

General Options

Frequency 1: 1

Sample size 1: 2

Frequency 2: 1

Sample size 2: 2

Range:

Sheet ☒ Workbook

☒ z test

Data format:

☒ Frequencies ☐ Proportions

☐ Monte Carlo method:

Number of simulations: 5000

OK Cancel Help

OPTIONS TAB

Choose the **Alternative hypothesis**: to be used for the test. Enter the value of the supposed difference between the proportions of the sample in the **Hypothesized ratio (D)**: option. Enter the significance level for the tests (default value: 5%) as the **Significance level (%)**:. Choose the **Variance**: method used to estimate the variance of the difference between the proportions. Select the radio button that corresponds to the correct formula.

Tests for two proportions

General Options

Alternative hypothesis:

Proportion 1 - Proportion 2 $\neq$ D

Hypothesized difference (D): 0

Significance level (%): 5

Variance:

☒ $p1q1/n1+p2q2/n2$

☐ $pq(1/n1+1/n2)$

OK Cancel Help

MULTINOMIAL GOODNESS OF FIT TEST

Use this tool to test whether the observed frequencies of the values (categories) of a qualitative variable correspond to the expected frequencies or proportions. The test is based on the multinomial distribution which is the extension of the binomial distribution when there are more than two possible outcomes. Let k be the number of possible values (categories) for variable X . We write $p_1, p_2, \dots, p_k$ the probabilities (or densities) corresponding to each value. Let $n_1, n_2, n_3, \dots, n_k$ be the frequencies of each value for a sample.

The null hypothesis of the test writes:

H_0 : The distribution of the values in the sample is consistent with what is expected, meaning the distribution of the sample is not different from the distribution of X .

The alternative hypothesis of the test writes:

H_a : The distribution of the values in the sample is not consistent with what is expected, meaning the distribution of the sample is different from the distribution of X .

XLSTAT offers the Chi-square test and the Monte Carlo test as alternatives for generating the test statistic

MULTINOMIAL GOODNESS OF FIT TEST TAB

In the **Data Format:** options activate the option indicating whether you will choose expected *Frequencies* or expected *Proportions*. In the **Frequencies:** option select the data corresponding to the observed frequencies in the Excel worksheet. In the **Expected proportions:** option select the data corresponding to the expected frequencies or to the expected proportions. If you select expected frequencies, they must sum to the same value as the sum of the observed frequencies. Activate the *Column labels* option if the first line of the data selected contain a label. Activate the **Chi-square test:** option to use the Chi-Square test. Activate the **Monte Carlo method:** to use the simulation method and enter the number of simulations. Enter the significance level for the tests (default value: 5%) as the **Significance level (%)**.